

Ad Copilot — Metrics & Attribution Spec

Status: Draft v1.0 **Owner:** Product, in partnership with Data/Analytics **Related docs:** Case Study Narrative · PRD · User Stories · AI Workflow Design · Architecture Doc (next) · Release Plan (next)

This doc defines the performance marketing metrics Ad Copilot depends on — not as marketing vocabulary, but as precisely-defined product signals with sourcing, calculation, latency, and known limitations. Every drift-detection rule and confidence input in the AI Workflow Design doc traces back to a definition here.

1. Why This Doc Exists

The AI Workflow Design doc references “target CPA,” “signal strength,” and “data volume” as inputs to confidence scoring. None of that is trustworthy unless the underlying metrics are defined precisely and their limitations are known upfront — performance marketers will stop trusting the product the first time its numbers don’t reconcile with what they see in Ads Manager.

2. Core Metrics Used as Product Signals

Metric	Definition	Source	Used for
ROAS (Return on Ad Spend)	Attributed revenue ÷ ad spend, over a defined window	Platform-native (Meta/Google conversion + revenue data)	Primary drift-detection target for e-commerce accounts; north-star input
CPA (Cost per Acquisition)	Ad spend ÷ number of attributed conversions	Platform-native	Primary drift-detection target for lead-gen accounts
CPL (Cost per Lead)	Ad spend ÷ number of attributed leads (a	Platform-native, mapped from the account’s defined “lead” conversion	Secondary target for lead-gen accounts running multi-stage funnels

	conversion subtype)	event	
CPC (Cost per Click)	Ad spend ÷ clicks	Platform-native	Early-signal input — CPC shifts often precede CPA/ROAS shifts and are used as a leading indicator in drift detection, not a standalone target
CTR (Click-Through Rate)	Clicks ÷ impressions	Platform-native	Creative fatigue signal (§4) — a declining CTR trend on a stable audience is the primary trigger for the “generate replacement creative” workflow

Deliberate scope decision: all five metrics are sourced from the ad platforms’ own reporting, not recomputed independently. This is consistent with the PRD’s non-goal of building a custom attribution engine — the product’s credibility depends on its numbers matching what the marketer already sees natively.

3. Attribution Approach

v1 decision: platform-native attribution, not a custom multi-touch model.

- Meta and Google each report conversions using their own attribution windows and models (e.g., Meta’s default 7-day click / 1-day view, Google’s data-driven attribution where eligible).
- Ad Copilot ingests these platform-reported conversion numbers as-is and does **not** attempt to reconcile or override them with an independent model.
- **Why:** a custom attribution layer that disagrees with the platform’s own dashboard creates an immediate credibility problem (“why does your tool say something different from Meta?”) and roughly doubles engineering scope for a 0-to-1 build. This tradeoff is explicit and revisited only if customer research in a later phase shows attribution disagreement is a top blocker.
- **Known limitation this creates:** cross-platform conversion counting can double-count a single customer journey that touched both Meta and Google ads. v1 does not attempt de-duplication across platforms; this is disclosed in-product (a “platform-reported, not deduplicated across channels” tooltip) rather than silently presented as a unified truth.

4. Conversion Tracking Architecture (product-level, not full

technical spec)

- Ad Copilot relies on the advertiser's own existing conversion tracking setup (Meta Pixel/Conversions API, Google Ads conversion tracking/Enhanced Conversions) rather than requiring a new tracking implementation.
- **Data freshness:** performance data is pulled hourly (per PRD NFR-1); conversion data itself may still be subject to the platform's own attribution lag (conversions can be attributed retroactively up to the platform's attribution window, meaning yesterday's ROAS figure can still shift for several days after). Drift detection accounts for this by weighting the most recent 24–48 hours of data less heavily than the preceding week when calculating trend stability (tie-in to "signal strength" in the AI Workflow Design doc, §4).
- **Known data-quality risks surfaced to the user, not hidden:**
 - **iOS/App Tracking Transparency (ATT) impact** — accounts with a high share of iOS conversions may see modeled/estimated conversions from the platform rather than fully observed ones. Ad Copilot surfaces the platform's own modeled-vs-observed flag where available rather than treating all conversions as equally certain.
 - **View-through vs. click-through conversions** — these are reported separately where the platform allows, since blending them can make CPA look artificially strong. Drift-detection targets let the marketer choose whether to include view-through conversions in their target definition.
 - **Sampling/reporting delay in platform APIs** — Meta and Google APIs occasionally report provisional numbers that adjust within 24–72 hours. Any recommendation based on data still within this provisional window is annotated as such in its reasoning (tie-in to AI Workflow Design §2's "no unverifiable claims" prompt constraint — the model isn't allowed to present provisional data with full confidence language).

5. How Metrics Feed the AI System

Direct references back to the AI Workflow Design doc:

- **Drift detection (§6 in that doc):** a campaign's live metric (ROAS or CPA, per account type) is compared to its target over a rolling window; the size and duration of the deviation is what generates a recommendation.
- **Confidence scoring's "signal strength" term:** derived from how far the deviation is from target and how stable it's been — a one-hour CPC spike is weighted very differently from a three-day CTR decline.

- **Confidence scoring's "data volume" term:** directly pulls from conversion counts and lookback window length defined here (the <100-conversions / <7-days conservative-confidence rule).
- **Creative fatigue detection (triggers US-4, scheduled refresh):** CTR trend decline on an otherwise-stable audience/budget is the specific signal used, since CTR is a faster-moving, lower-noise leading indicator than downstream conversion metrics.

6. North-Star Metric (Product-Level, Distinct from Marketing Metrics Above)

Incremental ROAS improvement attributable to accepted AI suggestions.

- Calculated by comparing performance of campaigns in the period after a suggestion was accepted against a trailing baseline for that same campaign (not against a control group in v1 — a true holdout/control methodology is a valid v2 investment once volume supports it).
- Deliberately excludes campaigns where no suggestion was accepted, to avoid diluting the signal with unrelated account-level performance noise.
- This metric is reported to the Marketing Lead persona (PRD §5) as the primary product-value proxy, separate from the guardrail metrics below.

7. Guardrail Metrics (Not Growth Metrics)

- **LLM cost as % of managed ad spend** — must stay under a fixed ceiling per account; a rising trend here is treated as a product-health alarm, not optimized against.
- **Suggestion acceptance rate** — tracked as a leading indicator of trust (per PRD goals), but deliberately not gamed by making suggestions "easier to accept" (e.g., trivially small budget nudges) — acceptance rate is reviewed alongside actual incremental ROAS to catch that failure mode.
- **Escalation rate** — the % of AI actions landing in the Escalate lane; expected to decrease over time as account trust tiers mature, but a sudden drop is investigated as a possible threshold-miscalibration risk rather than celebrated uncritically.

8. Explicit Non-Goals of This Spec

- Does not define a custom attribution model (per §3).

- Does not define incrementality/lift testing methodology in v1 (flagged as a valid v2 investment in §6).
- Does not attempt cross-platform conversion deduplication (per §3), though this limitation is disclosed in-product.